Appendix 2: Lake Washington Analysis

Here we show the GAM and DLM analysis for the Lake Washington dataset. For complete code, see the full quarto document. We include some code snippets here for reference.

Data Processing

In Lake WA the major change ecosystem change is related to sewage in the lake – adding wastewater plants in the 1960s reduced phosphorous (Figure S2.1) and in turn, blue-green algae (Figure S2.2) which can be seen by examining time series plots of the data.

We can then use the MARSS package to fit a DLM. We'll filter the data to only use counts before 1974, and log transform densities. We also z-score the covariate. This is a good example of imperfect data – there's a couple of missing observations in the covariate, which we can't have in a DLM or GAM – so let's just fill them in with a spline. We can do this as follows:

```
dat <- dplyr::filter(dat, Year < 1974) %>%
  dplyr::mutate(y = log(Bluegreens), zp = as.numeric(scale(TP)))

interpolated <- spline(dat$zp, xout = 1:nrow(dat))$y
dat$zp[which(is.na(dat$zp))] <- interpolated[which(is.na(dat$zp))]
```

The response data exhibit a strong seasonal cycle – and it's a good idea to de-season that to focus on the phosphorous effect. Here we de-season the data:

```
# Fill in a few missing values
dat$y_interp <- dat$y
interpolated <- spline(dat$y, xout = 1:nrow(dat))$y
missing <- which(is.na(dat$y_interp))
dat$y_interp[missing] <- interpolated[missing]

dat <- dplyr::group_by(dat, Month) %>%
  dplyr::mutate(month_mean = mean(y,na.rm=T)) %>%
```

```
  dplyr::ungroup() %>%
  dplyr::mutate(y_adj = y_interp - month_mean)
```

Last, we should look at the lags between driver and response relationships which we can do using a CCF (Figure S2.3). The CCF here indicates a 2-4 month lag, where TP and algae are more strongly associated with one another.

We'll use a lag of 4 months, which we can apply as follows:

```
lag_n <- 5
dat$lagged_zp <- c(rep(NA, lag_n), dat$zp[1:(nrow(dat)-lag_n)])
dat <- dat[-c(1:lag_n),]
```

DLMs

First, we need to define the model – this block is basically copied from the MARSS book, the salmon survival case study

```
m <- 2
TT <- nrow(dat)
B <- diag(m) ## 2x2; Identity
U <- matrix(0, nrow = m, ncol = 1) ## 2x1; both elements = 0
Q <- matrix(list(0), m, m) ## 2x2; all 0 for now
diag(Q)[1] <- 0.0000001 ## time-independent intercept
diag(Q)[2] <- c("q.beta") ## time-dependent slope
#diag(Q) <- c("q.alpha", "q.beta") ## 2x2; diag = (q1,q2)
Z <- array(NA, c(1, m, TT)) ## NxMxT; empty for now
Z[1, 1, ] <- rep(1, TT) ## Nx1; 1's for intercept
Z[1, 2, ] <- dat$lagged_zp ## Nx1; predictor variable
A <- matrix(0) ## 1x1; scalar = 0
R <- matrix("r") ## 1x1; scalar = r
x0 = matrix(c(0, 0)) ## initial conditions known
## only need starting values for regr parameters
inits_list <- list(x0 = matrix( nrow = m))
## list of model matrices & vectors
mod_list <- list(B = B, U = U,x0 = x0, Q = Q, Z = Z, A = A, R = R)
## convert response to matrix
dat_mat <- matrix(dat$y_adj, nrow = 1)
## fit the model -- crank up the maxit to ensure convergence
dlm_1 <- MARSS(dat_mat, inits = inits_list, model = mod_list,
control = list(maxit=4000))
```

```
Success! abstol and log-log tests passed at 19 iterations.
Alert: conv.test.slope.tol is 0.5.
Test with smaller values (<0.1) to ensure convergence.

MARSS fit is
Estimation method: kem
Convergence test: conv.test.slope.tol = 0.5, abstol = 0.001
Estimation converged in 19 iterations.
Log-likelihood: -168.2105
AIC: 340.4209   AICc: 340.5091

         Estimate
R.r         0.311
Q.q.beta    0.227
Initial states (x0) defined at t=0

Standard errors have not been calculated.
Use MARSSparamCIs to compute CIs and bias estimates.
```

Let's put the state estimates back into the original dataframe and do some plotting. The slope is generally positive, which is what we expect for the hypothesized relationship between bluegreen algae and total phosphorus – and exhibits clear variation through time (Figure S2.4). We can also plot the fitted values from our model (Figure S2.5).

And we can examine the residuals for a temporal trend, and see that because the DLM is flexible, there's not much evidence of a trend (Figure S2.6) and there are no issues wiht the q-q plot (Figure S2.7).

GAMs

We can fit a GAM model 5 ways, 1) so that the smoother is a function of phosphorus which represents a nonlinear functional relationship between total phosphorus and bluegreen algae that is not temporally structured thus the relationship is stationary and fixed through time. This is how GAMs are typically structured for ecological analyses. 2) a temporally structured model such that the abundance of bluegreen algae varies as a function of time unrelated to phsophorus, 3) A temporally structured model so that the smooth function of phosphorus varies by time, and 4) a smooth on date as a driver of bluegreen algae where the smooth through time varies by phosphorus level. Finally, these relationships can be fit as 5) a tensor product which provide a multivariate function spanning the dimensions of time and phosphorus. We will not discuss tensor products in greater detail and only include them for completeness. We show the model predicted bluegreen algae trend through time using these five modeling approaches (Figure S2.8). While they are quite similar, the predictions diverge sustantially at the end of the time series.

```
dat$n_date <- as.numeric(as.factor(dat$date))

gam_cr1 <- gam(y_adj ~ s(lagged_zp, bs = "cr"), data = dat) #1
gam_cr2 <- gam(y_adj ~ s(lagged_zp, by = n_date, bs = "cr"), data = dat) #2
gam_cr3 <- gam(y_adj ~ s(n_date, by = lagged_zp, bs = "cr"), data = dat) #3
gam_cr4 <- gam(y_adj ~ s(n_date), data = dat) #4
gam_cr5 <- gam(y_adj ~ te(lagged_zp, n_date), data = dat) #5
```

It is important to also examine the residuals from these models. For example, we can look at the residuals for the GAMs fit to the Lake Washington data. In looking across models, those that do not include time-dependent parameters show evidence of a persistent trend in the residuals that is not accounted for by the model (Figure S2.9) despite the q-q plot being reasonable for all model specifications (Figure S2.10).

```
resid_df_1 <- data.frame(date = dat$date, n_data=dat$n_date,resid = resid(gam_cr1),
                         Model = "s(Phos)")
resid_df_2 <- data.frame(date = dat$date, n_data=dat$n_date,resid = resid(gam_cr2),
                         Model = "s(Phos, by = date)")
resid_df_3 <- data.frame(date = dat$date, n_data=dat$n_date,resid = resid(gam_cr3),
                         Model = "s(date, by=Phos)")
resid_df_4 <- data.frame(date = dat$date,n_data=dat$n_date, resid = resid(gam_cr4),
                         Model = "s(date)")
resid_df_5 <- data.frame(date = dat$date, n_data=dat$n_date, resid = resid(gam_cr5),
                         Model = "te(Phos, date)")
resid <- rbind(resid_df_1, resid_df_2, resid_df_3, resid_df_4, resid_df_5)%>%
  left_join(p_values)

gam_resid<-ggplot(resid, aes(date, resid, group = Model)) +
  geom_point() +
  geom_smooth(method = "gam", formula = y ~ s(x)) + theme_classic() +
   geom_text(
    aes(x = as.Date('1965-11-01'), y = -3, label =paste0("p = ",round(p_value,4))),
    inherit.aes = FALSE)+
  facet_wrap(~ Model) +
  xlab("Date") +
  ylab("Residuals") +
  ggtitle("GAM Residuals")
```

There are a few more analysis approaches for running GAMs including the choice of smoother function (Figure S2.8 & Figure S2.11) and the number of knots, or the "wiggliness" (Figure

S2.12) of the smoothed relationship. We can plot these different results and include the results of the DLM for comparison, note that 'cr' and 'tp' are so similar they are difficult to discern from the plot. We can do a similar comparison for the choice of k. Here we compare a four different numbers of knots (Figure S2.12), we show an intentionally wide range of k (30 - 120) to illustrate the effects of k on estimated uncertainty, in practice k should be specified based on degrees of freedom and expectations of variability through time.

LM

Pollution reduction occurred over a multi-year period (1964 - 1968) and as such the change in the relationship between phosphorus and cyanobacteria is likely a continuous process evolving over time. Here we demonstrate a LM approach for the Lake Washington, choosing a year in the middle of the time period as the temporal break (1966). We find these results agree with our GAM/DLM results such that the best model (Table S2.1) includes a time-dependent slope where prior to 1966 there is a strong positive relationship between total phosphorus and cyanobacteria prior to 1966 and this relationship is reduced to nearly 0 after 1966 (Table S2.2). Given expected change in the relationship is supposed to be gradual through time, the hypothesized mechanism for the change in the relationship supports using a GAM or DLM rather than an LM but we include the LM to show agreement across the three model types for completeness.

Figures & Tables

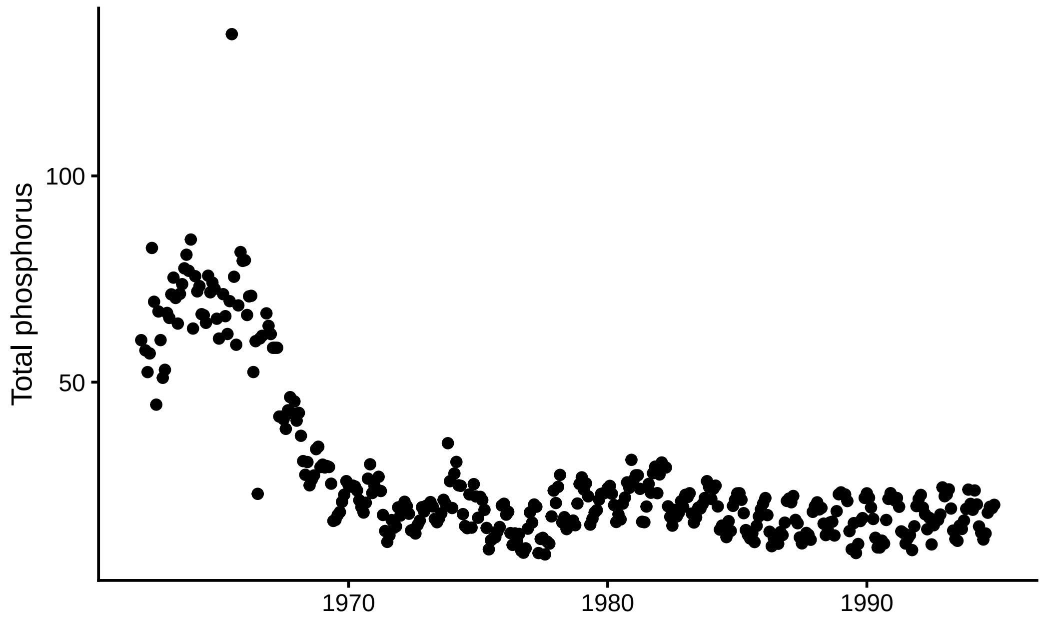

Figure S2.1: Time series of total phosphorus in Lake Washington

Figure S2.2: Time series of bluegreen algae (cyanobacteria) abundance in Lake Washington

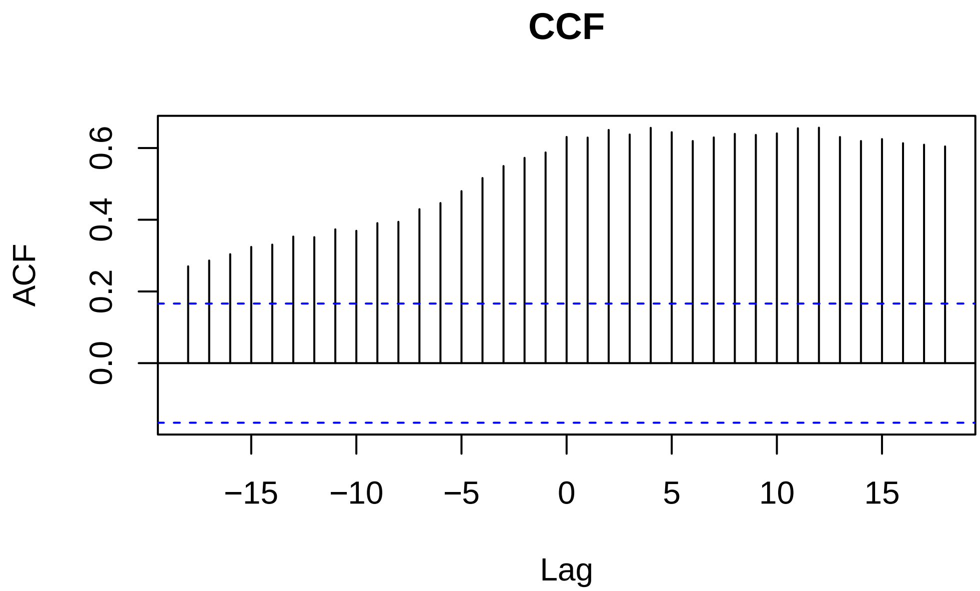

Figure S2.3: A CCF (Cross-Correlation Function) plot that represents the correlation between total phosphorus and blugreen algae abundance as a function of the lag (time difference) between them.

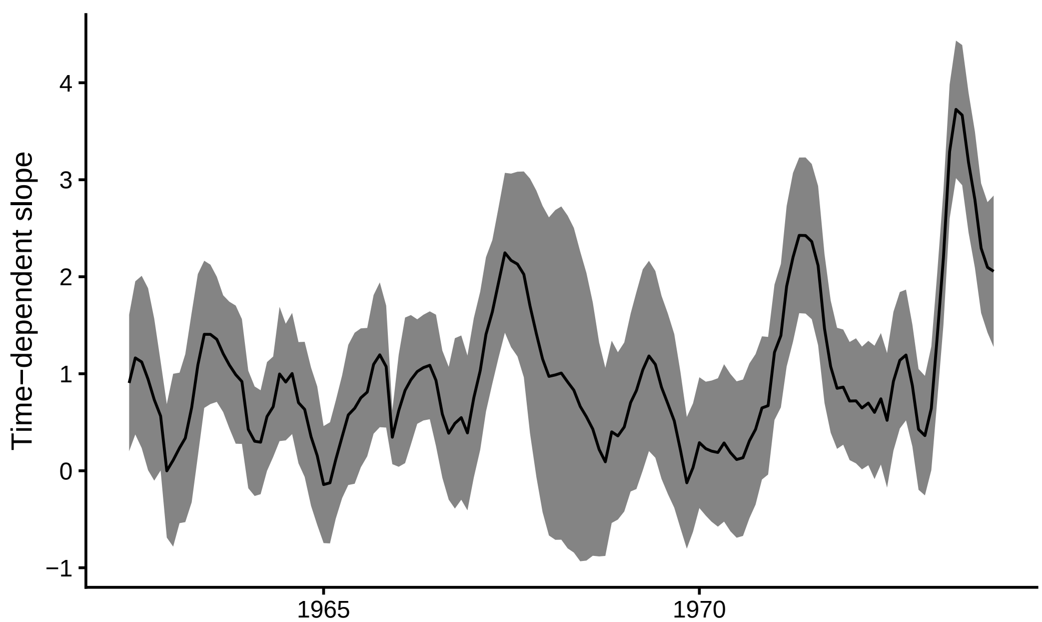

Figure S2.4: Time-dependent slope of the relationship between bluegreen algae (response) and total phosphorus (driver) from a DLM

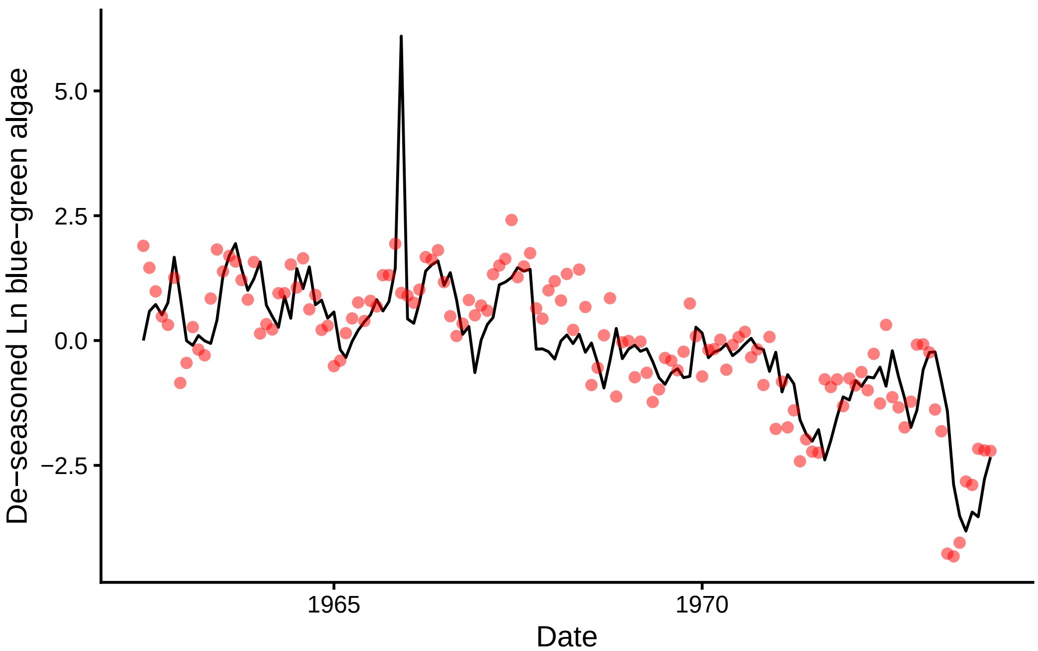

Figure S2.5: Fit of the DLM model with time-dependent slope (black line) to observed blue-green algae abundance (red points).

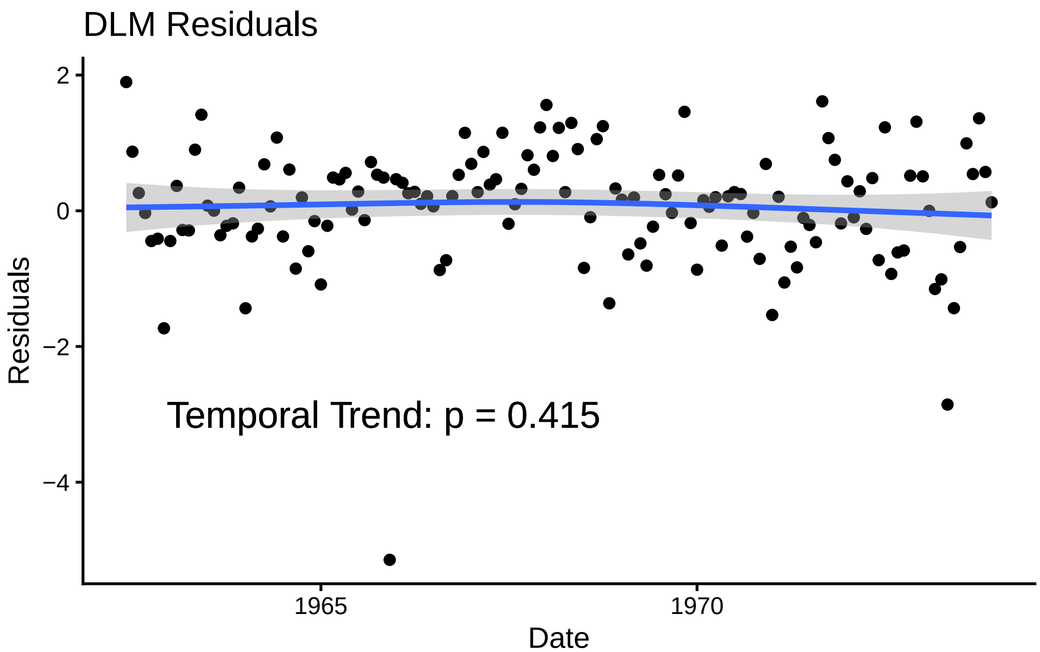

Figure S2.6: Time series of residuals from the DLM model fit with a time-dependent slope between blue-green algae and phosphorus. p-value denotes the significance of the trend in residuals through time.

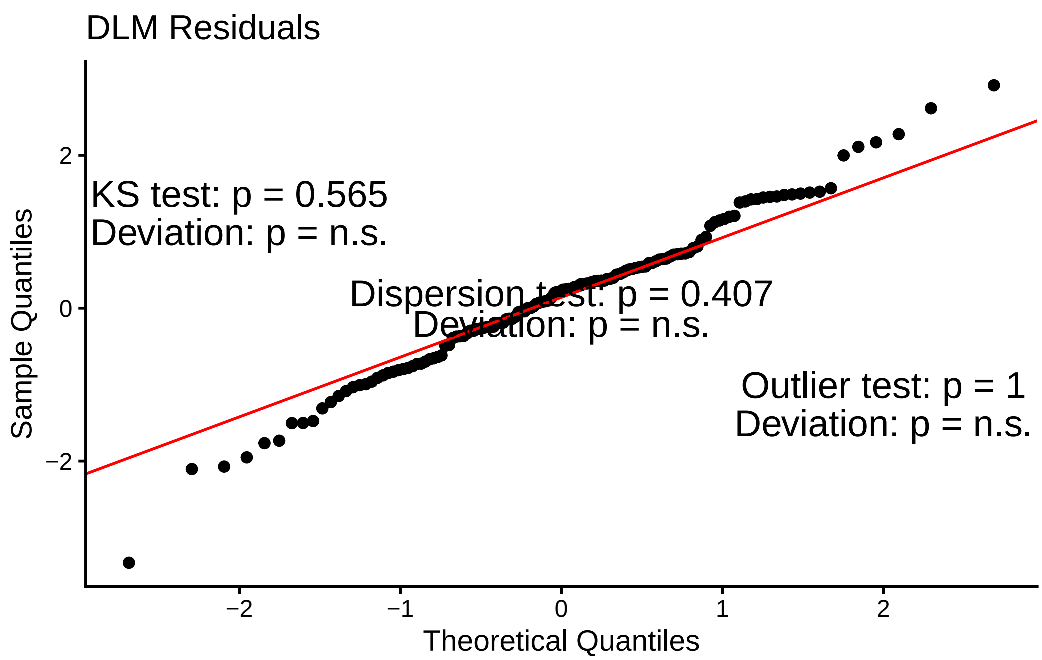

Figure S2.7: Q-Q plot from the DLM model fit with a time-dependent slope between blue-green algae and phosphorus

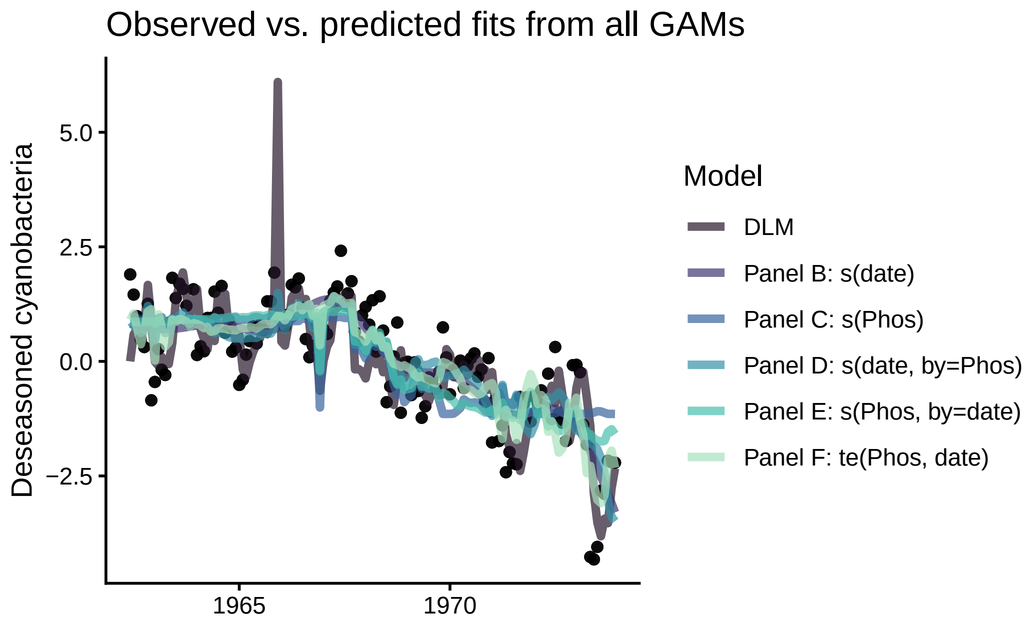

Figure S2.8: Fit of the GAM models (lines) to observed bluegreen algae (cyanobacteria) abundance (black points). Colors denote different specifications of the smoother functions.

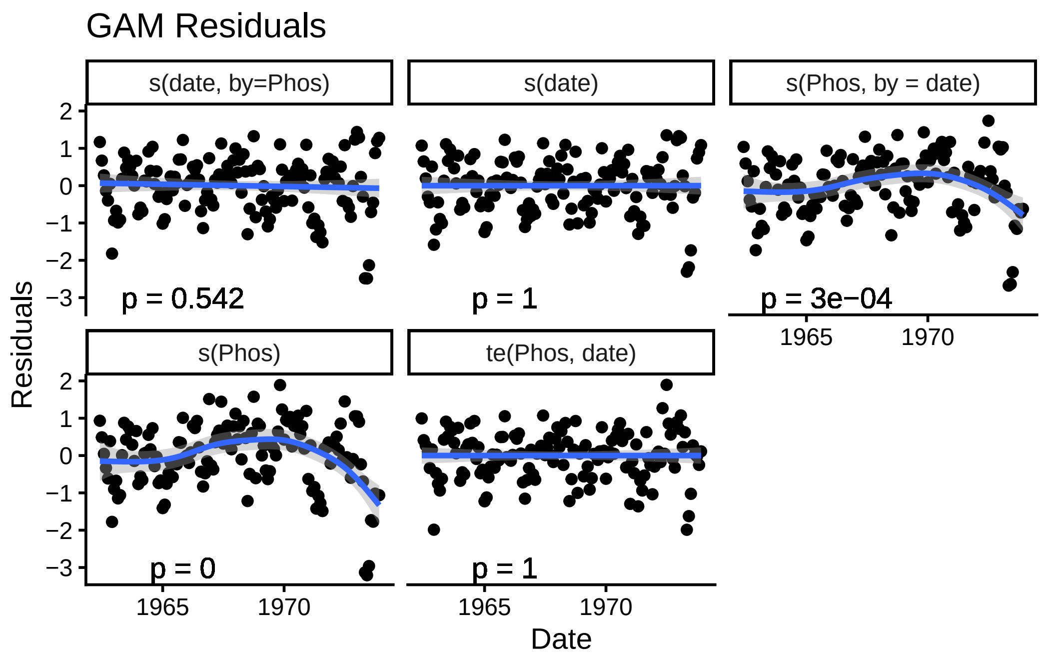

Figure S2.9: Time series of residuals from the five GAM models where residuals from models that do not have a temporally structured smooth term (s(Phos) and s(Phos, by = date)) have a trend through time. p-value denotes the significance of the trend in residuals through time.

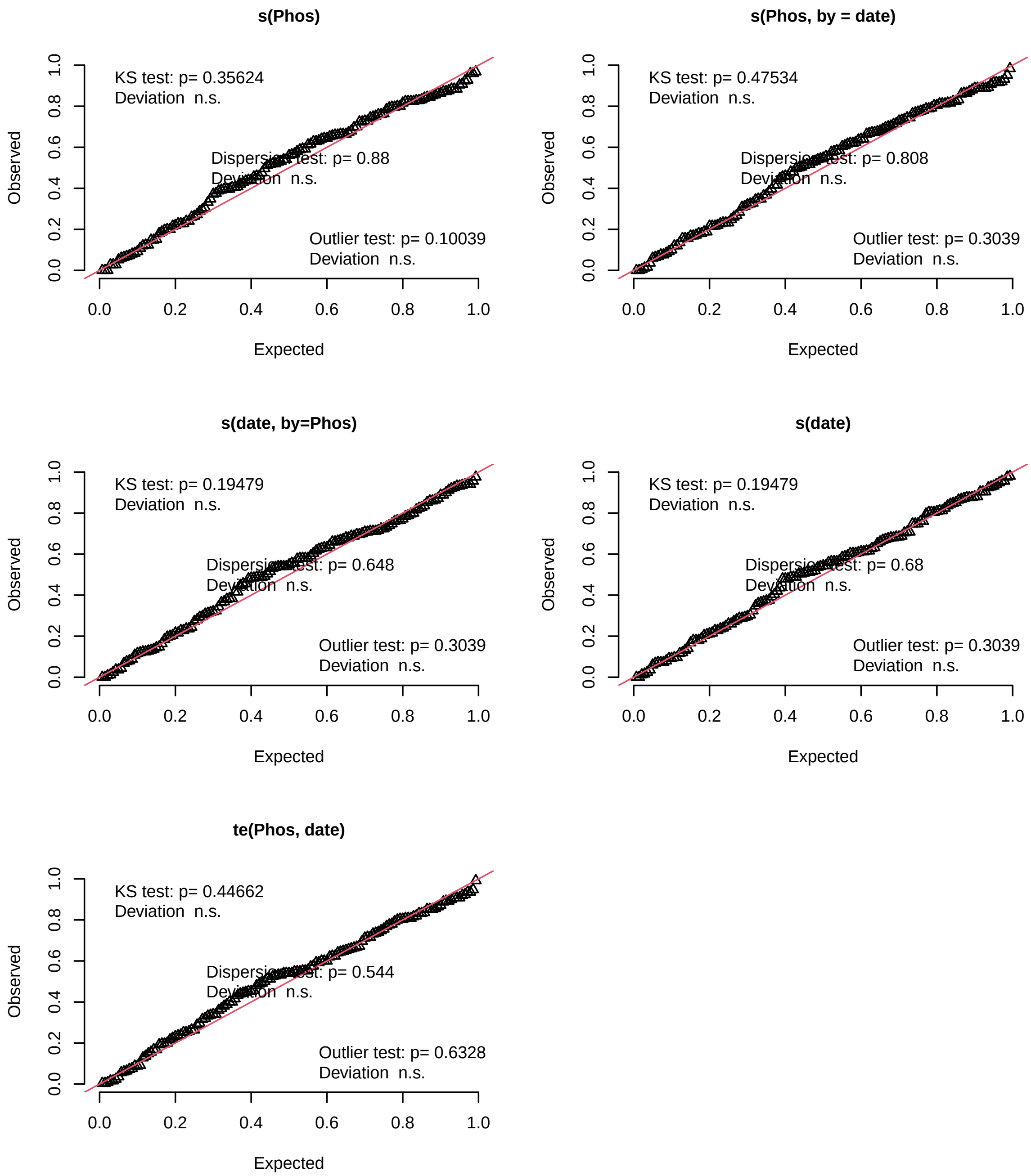

Figure S2.10: Q-Q for each of the five GAM models where residuals from models that do not have a temporally structured smooth term (s(Phos) and s(Phos, by = date)) have a trend through time but no significant issues with overdispersion, outliers, or deviations from the 1:1 line.

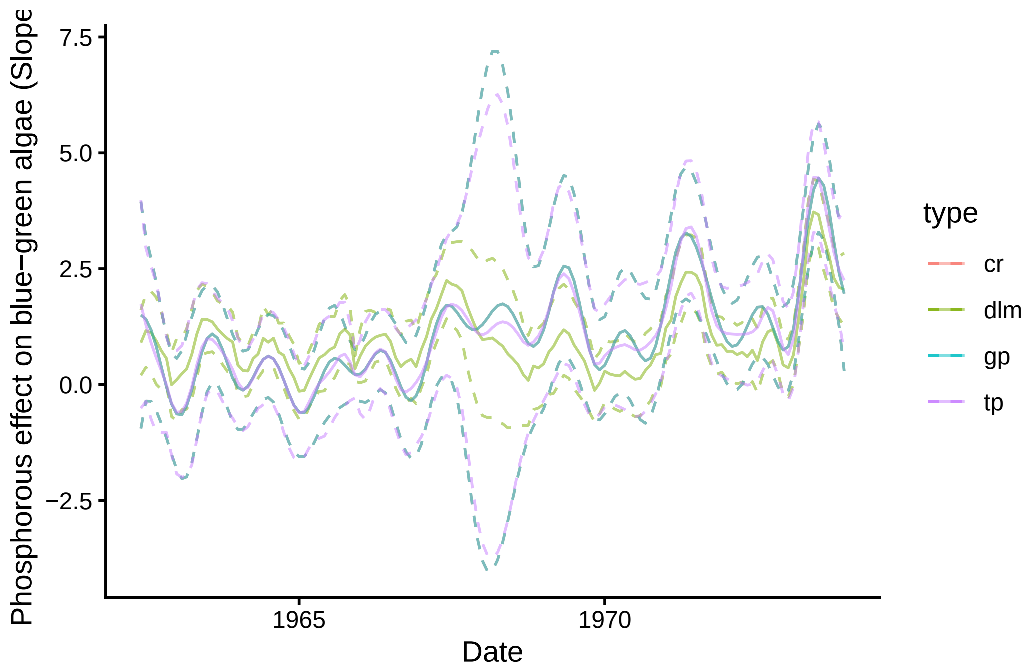

Figure S2.11: Effect of the choice of smoother function on estimates of time-dependent slope parameter where color denotes model type. 3 GAM smoother types (cr, tp, gp) are contrasted with a DLM (dlm).

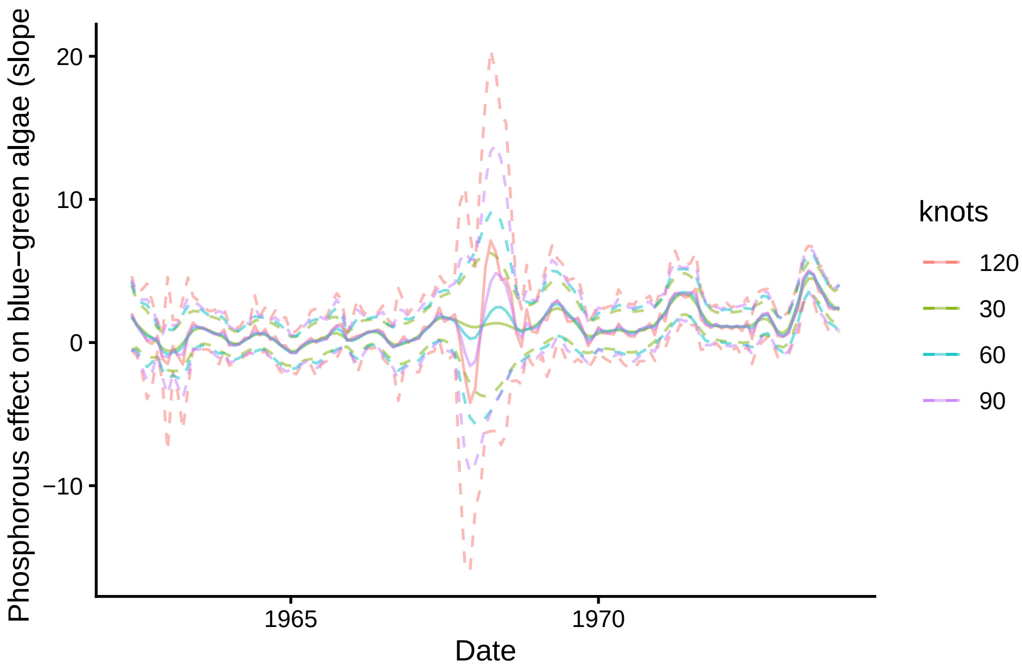

Figure S2.12: Effect of smoothing degrees of freedom on the standard errors of the estimates of time-dependent slope parameter where color denotes model type. We show 4 distinct knots (colors)

Model	AIC
Time-Independent Intercept	469.15
Time-Dependent Intercept	388.96
Time-Independent Slope	430.12
Time-Dependent Slope	372.37

Table S2.1: AIC values comparing linear models with time-dependent abundance models versus nonstationary relationship models

term	Estimate	Std. Error	t value	Pr(>$\lvert t \rvert$)
(Intercept)	0.35	0.14	2.47	0.01
lagged_zp	1.45	0.17	8.62	0.00
as.factor(period)Treatment	0.21	0.29	0.71	0.48
lagged_zp:as.factor(period)Treatment	-1.22	0.27	-4.52	0.00

Table S2.2: Model coefficients for the nonstationary relationship model between total phosphorus and cyanobacteria